A Comment on "Does Forest Loss Increase Human Disease? Evidence from Nigeria"*

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Abstract

Berazneva and Byker (2017) estimate the effect of forest loss on malaria, diarrhea, and respiratory infections among children under five in Nigeria, combining satellite data on forest change with the 2008 and 2013 Demographic and Health Surveys (DHS). They find that forest loss increases malaria incidence in the first two years after the loss, but has no effect on diarrhea or cough. Using the replication package, we reproduce the published estimates almost exactly, and they are robust to updated survey weights and to a reduced set of controls. Reconstructing the analysis data from raw sources yields a forest loss measure with a different spatial distribution. With these data, the cumulative effect on malaria is positive and of similar size in the first two years, but it is imprecisely estimated and does not fade afterwards. Adding the 2018 DHS wave yields a smaller effect with a similar rise and decline. For diarrhea, the reconstructed data point to a delayed positive effect; for cough, we find no consistent relationship. Overall, the direction of the malaria effect holds up, but its timing and statistical significance depend on how forest loss is measured.

KEYWORDS: Replication, Deforestation, Malaria, Child Health, Environmental Degradation, Nigeria

JEL CODES: I12, I15, J13, O13, O15, Q23

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1 Introduction

Berazneva and Byker (2017) study whether forest loss harms the health of young children in Nigeria. They focus on three illnesses that account for a large share of deaths among children under five in the country: malaria (proxied by fever), diarrhea, and acute respiratory infections (proxied by cough). By linking a high-resolution global dataset of forest change (Hansen et al. 2013) to geo-referenced child-level data from the 2008 and 2013 Nigeria Demographic and Health Surveys (DHS), the study provides new causal evidence on the health costs of forest loss in sub-Saharan Africa and contributes to a small but growing literature on the health consequences of natural resource degradation.

The identification strategy exploits variation in forest loss over time within local government areas (LGAs), Nigeria's second-smallest administrative units. For each DHS cluster, the study computes the share of pixels in a buffer zone around the cluster that experienced forest loss in the survey year and in each of the three preceding years. It then relates the incidence of each illness in the two weeks before the interview to these measures, controlling for LGA and survey-year fixed effects, region-specific seasonality, and a rich set of child, household, and cluster characteristics, including soil fertility, altitude, and changes in nighttime lights. In the most complete specification, a loss of 1 percent of forest cover in the previous year increases malaria incidence by about two percentage points; a one-standard-deviation increase in forest loss raises malaria incidence by about 4.5 percent of its mean. The dynamic pattern, with positive effects in the first two years after forest loss that subsequently fade, is consistent with a temporary ecological disturbance as described in the tropical medicine literature. The study finds no significant effects on diarrhea or cough.

In this comment, we assess the computational reproducibility and robustness of these findings in three steps. First, we re-estimate the main specifications using the analysis dataset and code in the replication package. Second, following the documentation, we reconstruct the analysis dataset from the original raw data sources and re-estimate the models. Third, we conduct two robustness checks: we extend the panel with the 2018 DHS wave, and we estimate a reduced specification with fewer and less collinear controls.

The analysis data and code reproduce the published results almost exactly. We note minor discrepancies in the figures used to de-normalize the survey weights; updating them leaves the estimates essentially unchanged. The original estimates are also robust to a reduced set of controls.

Reconstructing the analysis dataset from raw sources is only partly successful. The survey-based variables match the original data closely, but the spatial variables

differ significantly, most notably forest loss: in our reconstruction, forest loss is concentrated in southern Nigeria, whereas the original analysis data show forest loss distributed more evenly across the country. Since the DHS cluster locations coincide in both datasets, these differences most likely arise in the processing of the geospatial data, for which no code is included in the replication package. Our reconstruction also deviates from the documented procedure in a few respects, such as the buffer around urban clusters and the LGA boundaries.

With the reconstructed data, the estimated effect of forest loss on malaria remains positive. Cumulated over the first two years after forest loss, it is of similar size as in the original study (3.2 compared with 3.5 percentage points for a loss of 1 percent of forest cover), but it is less precisely estimated: none of the cumulative effects is statistically significant at the 5 percent level. Unlike in the original study, the effect also does not fade after three years. Adding the 2018 DHS wave yields a smaller cumulative effect of 2.2 percentage points two years after forest loss, which then declines, similar to the pattern in the original study. For diarrhea, the reconstructed data show a positive effect that builds up over three years, which the original data do not show; for cough, isolated coefficients are statistically significant, but their signs differ across datasets.

Taken together, our results confirm that the published findings are computationally reproducible from the analysis data and robust to the choice of controls. With the reconstructed data, the direction of the malaria effect holds up, but its timing and statistical significance depend on how the forest loss measure is constructed. Because the code for the original geospatial processing is not available, we cannot determine which of the differences in processing drives these results.

The remainder of this comment is structured as follows. Section 2 assesses computational reproducibility from the analysis data and from raw sources. Section 3 presents the robustness checks, and Section 4 concludes.

2 Computational Reproducibility

2.1 Replication Package

Berazneva and Byker (2017) provide a replication package that contains the final analysis dataset, the code that produces the main results, and a README that lists the raw data sources and describes the main data preparation steps. Raw data and data-cleaning code are not included. Table 1 summarizes the contents of the package and our assessment of its reproducibility. The published results are fully reproducible from the analysis data. Reproduction from raw data is possible only in part, because several links to the raw data in the README are no longer active and no code for the geospatial processing is included.

2.2 Reproduction Using the Analysis Data

The main specification of Berazneva and Byker (2017) is

$$Y_{itc} = \alpha_0 + \sum_{j=0}^{3} \beta_j \, loss_{tc}^{j} + \mathbf{X}'_{itc}\gamma + \mathbf{LGA}'_c\pi + \mu' \, month_{itc} + \mathbf{DHSyear}'_t\theta + \sum_{m=1}^{12} \lambda_m \, month_{tc}^{m} \times Region_c \times DHSyear_t + \epsilon_{itc}, \quad (1)$$

where Y_{itc} indicates whether child i, surveyed in DHS wave t in cluster c, had the respective illness in the two weeks before the interview; $loss_{tc}^{j}$ is the share of pixels within the buffer around cluster c that experienced forest loss j years before the survey; $\mathbf{X}_{itc}$ is a vector of child, household, and cluster characteristics, including soil fertility, altitude, tree cover in 2000, and changes in nighttime lights; $\mathbf{LGA}_c$ and $\mathbf{DHSyear}_t$ denote LGA and survey-wave fixed effects; $month_{itc}$ is the month of the interview; and the interactions of month, region, and survey wave capture region-specific seasonality. Standard errors are clustered at the LGA level, and the estimation accounts for the DHS sampling design. Since forest loss is measured as a share, a coefficient of 2 implies that a loss of 1 percent of forest cover increases disease incidence by 2 percentage points.

Using the analysis dataset and code, we reproduce Table 1 of the original study. Column (1) of Tables 4–6 reports our estimates of the most complete specification for each outcome (columns 3, 6, and 9 of the original Table 1). For malaria, the coefficient on forest loss one year before the survey is 2.035 (standard error 0.956), compared with 2.037 (0.957) in the original study. Across all lags and outcomes, our estimates differ from the published values by at most 0.03. Panel (a) of Figure 2 reproduces Figure 2 of the original study, which plots the cumulative effect of forest loss, and adds the corresponding results for cough. As in the original study, the cumulative effect on malaria is positive and statistically significant one and two years after forest loss (2.5 and 3.5 percentage points) and declines to zero after three years.

Survey weights. When pooling DHS waves, the original study de-normalizes the survey weights so that each wave is weighted by the size of the population it represents, which is good practice. The population figures for women of reproductive age (WRA, aged 15–49) used in the replication code stem from an earlier revision of the UN population estimates, and the source cited in the code is no longer accessible. In addition, the number of interviewed women used for 2013 differs slightly from the figure in the DHS final report. We therefore update these figures (Appendix A). The update leaves the point estimates and standard errors essentially unchanged,

and the small differences between column (1) and the published estimates reflect it. We use the updated weights in all estimations reported in Tables 4–6.

2.3 Reconstruction of the Analysis Data from Raw Sources

To assess reproducibility from raw data, we rebuild the analysis dataset following the README and the description in the paper. The reconstruction combines four sources: the 2008 and 2013 Nigeria DHS, the Global Forest Change dataset (Hansen et al. 2013), the African SoilGrids soil property maps (Hengl et al. 2015), and the DMSP-OLS Nighttime Lights Time Series (US Air Force Weather Agency 2009). All data are aggregated to DHS clusters and LGAs as described below.

Health and household data. Child health outcomes and controls come from the Nigeria DHS, nationally representative, geo-referenced cross-sectional surveys conducted in 2008 and 2013 (NPC and ICF, 2009, 2014), to which access was granted upon request. Mapping the outcome variables to the survey questionnaires was straightforward, while identifying some of the control variables required more effort. All variables are available in the Children's Recode files, which we link to the GPS coordinates of the survey clusters; the coordinates are also used to obtain altitude. For the robustness check in Section 3.1, we add the 2018 DHS wave (NPC and ICF, 2019).

Geolinking. To protect the confidentiality of respondents, the DHS randomly displaces cluster coordinates: urban clusters by up to 2 km and rural clusters by up to 5 km, with 1 percent of rural clusters displaced by up to 10 km (Perez-Haydrich et al. 2013). The original study accounts for this displacement by drawing a 5 km buffer around every cluster. We instead match the buffer to the maximum displacement, using 2 km around urban clusters and 5 km around rural clusters, and extract the values of the forest loss, soil, and nighttime lights rasters within each buffer. All layers are reprojected from geographic coordinates (EPSG:4326) to a projected coordinate system in meters (EPSG:3857) for distance-based calculations. Raster values are averaged over the cells whose centre lies within the buffer and then across clusters within each LGA. To assign clusters to LGAs, the original study uses a shapefile of 710 LGAs from Africa Open Data, whereas we use the Subnational Administrative Boundaries for Nigeria (774 LGAs), published by the country office of the UN Office for the Coordination of Humanitarian Affairs (OCHA) on the Humanitarian Data Exchange (HDX).

Forest loss. The regressor of interest is derived from the Global Forest Change dataset (Hansen et al. 2013), which maps forest cover change based on Landsat

imagery at a resolution of one arc-second (about 30 meters at the equator). As in the original study, we use version 1.2 of the dataset, which covers the years 2000–2014. Forest loss is defined as a stand-replacement disturbance, that is, a change from forest to non-forest. We use the year-of-loss layer, which records for each pixel whether and in which year loss occurred, and the tree canopy cover layer for 2000, which records canopy closure for vegetation taller than 5 meters as a percentage. The loss measure captures gross loss and does not net out forest gain. The data are distributed in tiles, and Nigeria spans four of them; for buffers that straddle tile boundaries, we combine the pixels from all relevant tiles. For each cluster and year, forest loss is the share of pixels within the buffer that experienced loss in that year.

Soil fertility. Soil organic carbon, soil pH (measured in water), and cation exchange capacity are taken from the African SoilGrids maps at 250-meter resolution (Hengl et al. 2015). We extract values for the 0–5 cm and 5–15 cm depth intervals, average them to obtain values for the 0–15 cm layer, and then average these within each buffer and across clusters within each LGA.

Nighttime lights. As a proxy for economic activity (Chen and Nordhaus 2011), the study uses the stable nighttime lights composites of the DMSP-OLS Nighttime Lights Time Series (Version 4), produced by NOAA's National Geophysical Data Center (US Air Force Weather Agency 2009), at a resolution of 30 arc-seconds (about 1 km at the equator). We use the composites from satellite F16 for the years relevant to the 2008 wave and from satellite F18 for the years relevant to the 2013 wave, and aggregate them to buffers and LGAs as above.

2.4 Comparison with the Original Analysis Data

Following the original study, we restrict the sample to LGAs that contain at least one DHS cluster in each of the 2008 and 2013 waves. Table 2 compares the resulting dataset with the original analysis dataset, using two-sided t-tests for the equality of means and Kolmogorov–Smirnov tests for the equality of distributions.

The two datasets agree closely on the survey-based variables. Our reconstruction contains slightly more observations (for example, 41,737 compared with 41,409 children with a non-missing malaria outcome), but the means and distributions of the three outcomes are statistically indistinguishable. Both datasets contain 408 LGAs with valid identifiers, although they rely on different LGA boundary files; the 409th unit in the original analysis dataset corresponds to 40 observations without an LGA identifier, which we exclude. The DHS cluster locations coincide in both datasets.

By contrast, the distributions of all spatial variables differ significantly between the two datasets, as do the means of most of them. In magnitude, however, the differences in means are small: for the soil properties, for nighttime lights, for tree cover in 2000 (10.1 percent in our reconstruction compared with 10.8 percent in the original data), and for forest loss, whose means agree at the reported precision. The differences in forest loss lie mainly in its spatial distribution. Figure 1 illustrates this: in our reconstruction, forest loss between 2001 and 2012 is concentrated in southern Nigeria, in line with the location of the country's main forest areas, and limited in the north. The original analysis data, as shown in Figure 1 of Berazneva and Byker (2017), show substantial forest loss in many northern LGAs as well.

Because the cluster locations coincide, these differences most likely arise in the processing of the geospatial data. Our reconstruction departs from the procedure documented in the README in two respects: it uses a smaller buffer around urban clusters and different LGA boundaries. Without the original processing code, we cannot determine how much each of these choices contributes to the differences. In either case, the differences in the forest loss measure can be expected to affect the estimated relationship between forest loss and child health, to which we now turn.

2.5 Results Using the Reconstructed Data

Column (2) of Table 4 re-estimates the full specification using our reconstructed data, and panel (b) of Figure 2 shows the corresponding cumulative effects. The estimated effect of forest loss on malaria remains positive, but its timing differs from the original estimates. The coefficient on contemporaneous forest loss is 2.087 ($p = 0.095$), compared with 0.459 in column (1), whereas the coefficient on the one-year lag falls from 2.035 to 0.924 ($p = 0.435$). Cumulated over the first two years after forest loss, the effect amounts to 3.2 percentage points, close to the 3.5 percentage points implied by the original estimates. However, the cumulative effect does not decline after three years (4.1 percentage points), and none of the cumulative effects is statistically significant at the 5 percent level (p-values between 0.079 and 0.204). With the reconstructed data, forest loss is thus associated with higher malaria incidence of a magnitude similar to the original study, but the statistical evidence is weaker and the dynamic pattern of a temporary disturbance is not reproduced.

3 Robustness

3.1 Extending the Sample to the 2018 DHS Wave

The 2018 Nigeria DHS allows us to extend the panel by a third wave and to examine whether the results hold over a longer period. For this extension, we construct the forest loss measures for all three waves from version 1.7 of the Global Forest Change dataset, which covers the years 2000–2019 (Hansen et al. 2013), following the procedure described in Section 2.3, and we restrict the sample to LGAs observed in all three waves. Because the DMSP-OLS nighttime lights series ends in 2013, we omit the nighttime lights controls from the full specification in column (3); the reduced specification in column (6) excludes them anyway. Data from the successor sensor, VIIRS, are not directly comparable with DMSP-OLS. A harmonized DMSP–VIIRS series covering 1992–2018 is available (Li et al. 2020), but using it would require switching the nighttime lights source for all waves and thus depart further from the original study. Table 3 shows that the reported incidence of all three illnesses is higher in 2018 than in the earlier waves, particularly for fever (23.9 percent, compared with 15.8 and 13.5 percent in 2008 and 2013).

Column (3) of Table 4 and panel (c) of Figure 2 report the results, based on 55,245 children. The individual coefficients are smaller than with the 2008–2013 data, and none is statistically significant at the 5 percent level; the largest is the coefficient on forest loss two years before the survey (1.825, $p = 0.084$). The cumulative effect is positive at all horizons. It amounts to 2.2 percentage points two years after forest loss (95 percent confidence interval 0.19 to 4.30, $p = 0.032$) and declines to 0.4 percentage points after three years. The extended data thus show a pattern similar to the original study, with a cumulative effect that peaks two years after forest loss and then declines, although the effect is smaller and close to zero one year after forest loss.

3.2 Reduced Specification

The full specification includes a large number of controls, several of which are likely to be correlated with one another or with forest loss. Additional controls do not necessarily reduce bias and can reduce precision (Clarke 2005). To assess how sensitive the results are to the choice of controls, we estimate a reduced specification. First, we replace the month-by-region-by-survey-wave interactions and the month fixed effects with month-by-region interactions, as the former overlap substantially with the LGA and survey-wave fixed effects. Second, we drop the soil fertility and nighttime lights variables, which are closely related to drivers of forest loss such as agricultural expansion and economic development (Angelsen et al. 2014, Burgess et al. 2012).

Third, among the demographic controls, we retain the characteristics of the child, the education of the household head, and indicators of socioeconomic status, and drop variables that largely overlap with these, such as the ages of the household head and the respondent, housing characteristics, and the number of children ever born and living. Table A1 lists the variables included in each specification.

Columns (4)–(6) of Table 4 report the results. With the original analysis data (column 4), the estimates are nearly identical to those from the full specification: the coefficient on the one-year lag is 1.968 ($p = 0.040$), compared with 2.035 in column (1). The original finding is thus robust to the choice of controls. With the reconstructed data (column 5), the results again mirror those from the full specification, with a contemporaneous coefficient of 2.185 ($p = 0.080$). With the extended data (column 6), the coefficient on the one-year lag turns negative and statistically significant (-1.365, $p = 0.041$), while the remaining coefficients are positive but statistically insignificant. The R^2 changes little across specifications. For the original and the reconstructed data, the choice of controls thus matters little, which indicates that the differences between our estimates and the original ones stem mainly from the construction of the data.

3.3 Diarrhea and Respiratory Infections

Tables 5 and 6 report the corresponding results for diarrhea and cough, and Figure 2 shows the cumulative effects. With the original analysis data, we reproduce the null results of Berazneva and Byker (2017) for both outcomes, in the full as well as in the reduced specification (columns 1 and 4). With the reconstructed and extended data, some coefficients become statistically significant. For diarrhea, forest loss one year before the survey is associated with a higher incidence in the reconstructed data (2.248, $p = 0.021$; 2.145, $p = 0.022$, in the reduced specification), as is forest loss two years before the survey in the extended data (1.994, $p = 0.016$; 2.355, $p = 0.005$). The cumulative effect on diarrhea three years after forest loss is positive in both datasets (4.4 and 2.1 percentage points), with 95 percent confidence intervals that exclude zero in both datasets (0.80 to 7.92, $p = 0.017$, in the reconstructed data and 0.09 to 4.17, $p = 0.041$, in the extended data). For cough, the significant coefficients differ in sign across datasets: forest loss two years before the survey is associated with a lower incidence in the reconstructed data (-3.547, $p = 0.034$), whereas forest loss one year before the survey is associated with a higher incidence in the extended data (1.310, $p = 0.046$). None of the cumulative effects on cough is statistically significant. Given the number of coefficients estimated across lags, outcomes, and datasets, we interpret these isolated estimates with caution. Our results thus provide some indication of a delayed positive effect of forest loss on

diarrhea, which the original data do not show, and no consistent relationship for cough.

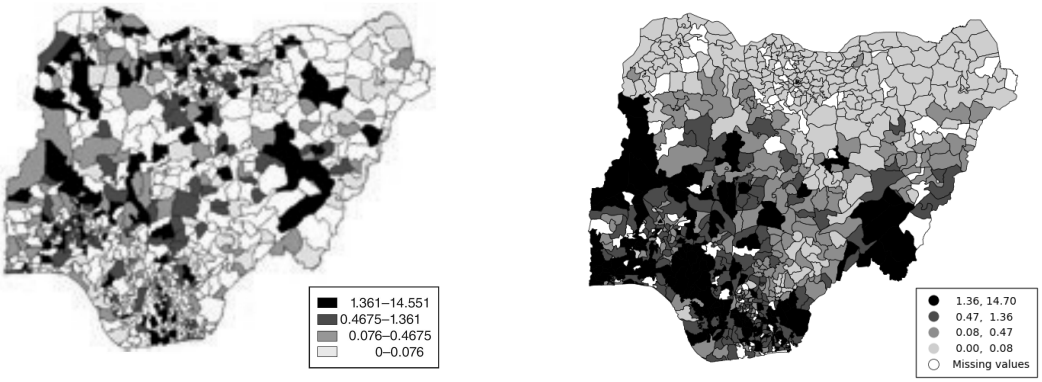
4 Conclusion

We assess the computational reproducibility and robustness of Berazneva and Byker (2017), who find that forest loss increases malaria incidence among young children in Nigeria. The replication package allows us to reproduce the published estimates almost exactly, and these estimates are robust to updated survey weights and to a reduced set of controls. Reconstructing the analysis dataset from the raw sources yields a forest loss measure with a different spatial distribution. With this measure, the estimated effect of forest loss on malaria remains positive and is of similar cumulative magnitude in the first two years after forest loss, but it is imprecisely estimated and does not fade afterwards. Extending the sample with the 2018 DHS wave yields a smaller effect that peaks two years after forest loss and then declines. For diarrhea, the reconstructed data point to a delayed positive effect that the original data do not show, while we find no consistent relationship for cough.

Our results therefore support the direction of the malaria effect reported in the original study, but not its precise timing and statistical significance, which depend on how the forest loss measure is constructed. Since the code for the original geospatial processing is not available, we cannot determine which of the differences in processing, such as buffer size or LGA boundaries, drives these results. This underlines the value of including the code for geospatial processing in replication packages, as choices about data versions, buffers, and aggregation can matter for the results.

Future work could explore heterogeneity in the effects of forest loss. As forest cover may need to be substantial to affect disease transmission (Laporta et al. 2021), one avenue is to focus on LGAs with substantial initial forest cover. Another is to restrict the sample to poorer or rural households, which tend to be more vulnerable, to test whether forest loss affects disease incidence differently across population groups (Bauhoff and Busch 2020).

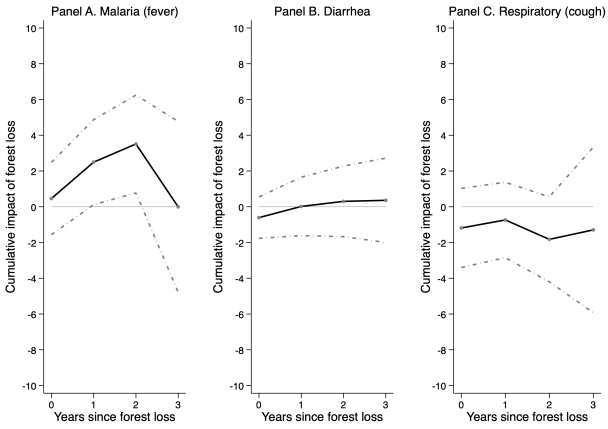
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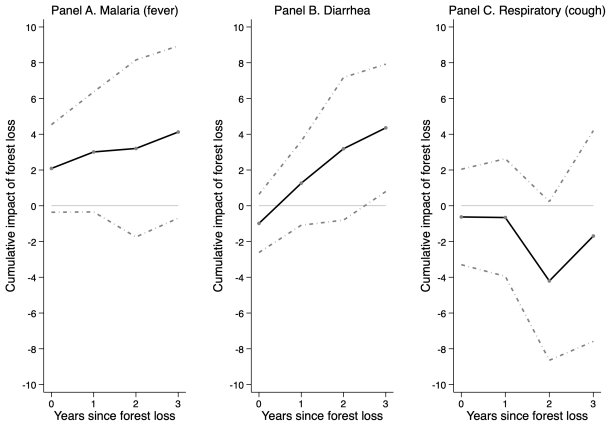
(a) Original (b) Reproduction

Figure 1: Geographic Variation of Forest Loss in Nigeria by LGA (2001–2012)

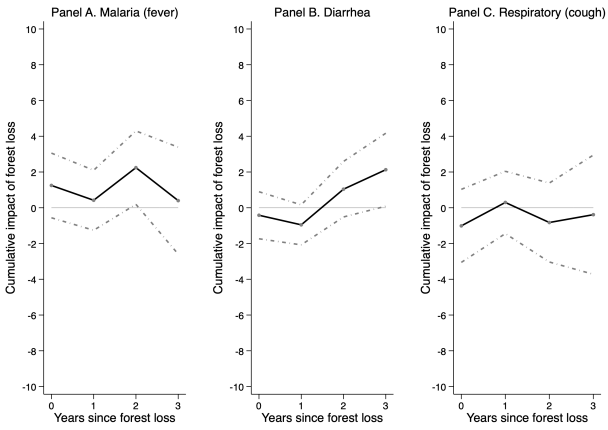
Notes: Panel (a) reproduces Figure 1 of Berazneva and Byker (2017), which shows the average share of pixels that lost tree cover between 2001 and 2012 within the buffers around DHS clusters, averaged across clusters in each of the 409 LGAs in their analysis sample. Panel (b) shows the same measure computed from our reconstructed data, using the buffers described in Section 2.3. In both panels, LGAs not observed in either the 2008 or the 2013 DHS wave are left unshaded.



(a) Original analysis data (2008 and 2013 DHS waves)



(b) Reconstructed data (2008 and 2013 DHS waves)



(c) Extended data (2008, 2013, and 2018 DHS waves)

Figure 2: Cumulative Impact of Forest Loss

Notes: Each panel plots the cumulative sum of the forest loss coefficients (β_0, $\beta_0+\beta_1$, $\beta_0+\beta_1+\beta_2$, $\beta_0+\ldots+\beta_3$) from the full specification; dash-dotted lines show 95 percent confidence intervals for each sum. Panel (a) reproduces Figure 2 of Berazneva and Byker (2017) with the original analysis data, code, and survey weights (their Table 1, columns 3, 6, and 9) and adds the results for cough. Panels (b) and (c) are based on columns (2) and (3) of Tables 4–6.

Tables

Table 1: Replication Package Contents and Reproducibility

Replication package item	**Fully**	**Partially**	**No**
Raw data provided			✓
Analysis data provided	✓		
Cleaning code provided			✓
Analysis code provided	✓		
Reproducible from raw data		✓	
Reproducible from analysis data	✓		

Notes: Contents of the replication package of Berazneva and Byker (2017) and our assessment of its reproducibility. The README lists links to the raw data sources, several of which are no longer active. Together with the absence of code for the geospatial processing, this allows only a partial reproduction from raw data.

Table 2: Descriptive Statistics of the Original and Reconstructed Datasets

	Original dataset			Reconstructed dataset			Test statistics	
Variable	Observations	Mean	SD	Observations	Mean	SD	t-stat	KS-stat
Malaria (fever)	41,409	0.144	0.351	41,737	0.144	0.352	0.135	0.000
Diarrhea	41,458	0.106	0.307	41,786	0.106	0.308	0.173	0.000
Respiratory (cough)	41,354	0.110	0.313	41,682	0.110	0.312	−0.111	0.000
Forest loss								
This year	46,654	0.001	0.004	47,039	0.001	0.003	−3.532***	0.047***
1 year ago	46,654	0.001	0.003	47,039	0.001	0.003	−0.680	0.039***
2 years ago	46,654	0.001	0.003	47,039	0.001	0.002	−5.256***	0.029***
3 years ago	46,654	0.001	0.002	47,039	0.001	0.002	−3.371***	0.032***
Change in luminosity								
This year	46,654	0.087	1.492	47,039	0.165	1.522	7.937***	0.083***
1 year ago	46,654	0.317	1.565	47,039	0.277	1.657	−3.802***	0.076***
2 years ago	46,654	−0.392	2.436	47,039	−0.312	2.440	5.035***	0.074***
Soil organic carbon	46,654	12.936	7.879	47,039	12.695	7.504	−4.788***	0.035***
Soil pH	46,654	59.801	3.855	47,039	59.882	3.695	3.271***	0.038***
Cation exchange capacity	46,654	10.472	4.236	47,039	10.452	3.907	−0.742	0.059***
Tree cover in 2000	46,654	10.820	13.360	47,039	10.134	12.836	−8.010***	0.041***

Notes: Descriptive statistics for the original analysis dataset and the dataset we reconstructed from raw sources following the documentation in the replication package. Both samples are restricted to LGAs observed in both the 2008 and the 2013 DHS waves. Forest loss is the share of pixels within the cluster buffer that lost forest cover in the indicated year. Soil pH is reported in the units of the source raster (pH × 10). The last two columns report two-sided t-tests for the equality of means and Kolmogorov–Smirnov tests for the equality of distributions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Incidence of Child Illnesses by DHS Wave

	Malaria (fever)			Diarrhea			Respiratory (cough)		
DHS wave	Observations	Mean	SD	Observations	Mean	SD	Observations	Mean	SD
2008	16,892	0.158	0.365	16,910	0.099	0.299	16,874	0.120	0.325
2013	22,237	0.135	0.342	22,260	0.109	0.312	22,200	0.105	0.306
2018	18,286	0.239	0.427	18,287	0.128	0.334	18,292	0.169	0.374
Total	57,415	0.175	0.380	57,457	0.112	0.316	57,366	0.129	0.336

Notes: Descriptive statistics for the three dependent variables in the extended dataset, which we reconstructed from raw sources and which covers the 2008, 2013, and 2018 DHS waves.

Table 4: Results for Malaria (Fever)

Malaria (fever)	Full specification			Reduced specification		
	Original (1)	Reconstructed (2)	Extended (3)	Original (4)	Reconstructed (5)	Extended (6)
Forest loss						
This year	0.459	2.087*	1.250	0.420	2.185*	0.994
	(1.034)	(1.249)	(0.922)	(1.009)	(1.246)	(0.951)
	[0.657]	[0.095]	[0.176]	[0.677]	[0.080]	[0.296]
1 year ago	2.035**	0.924	−0.830	1.968**	1.047	−1.365**
	(0.956)	(1.184)	(0.626)	(0.959)	(1.172)	(0.667)
	[0.033]	[0.435]	[0.185]	[0.040]	[0.372]	[0.041]
2 years ago	1.030	0.193	1.825*	1.168	−0.036	1.595
	(0.956)	(1.772)	(1.055)	(0.991)	(1.759)	(1.046)
	[0.281]	[0.913]	[0.084]	[0.239]	[0.984]	[0.128]
3 years ago	−3.540	0.920	−1.845	−3.210	0.587	−0.946
	(2.752)	(2.879)	(1.617)	(2.626)	(2.834)	(1.668)
	[0.199]	[0.749]	[0.254]	[0.222]	[0.836]	[0.570]
Observations	40,675	40,459	55,245	40,880	40,620	55,398
R^2	0.082	0.082	0.102	0.080	0.080	0.098

Notes: The dependent variable is the reported incidence of fever (malaria) in the two weeks preceding the survey among children under five. Columns (1)–(3) use the full specification of Berazneva and Byker (2017) (column 3 of their Table 1); columns (4)–(6) use the reduced specification described in Section 3.2, with the variables listed in Table A1. Columns (1) and (4) use the original analysis dataset, columns (2) and (5) the dataset reconstructed from raw sources (2008 and 2013 DHS waves), and columns (3) and (6) the extended dataset that adds the 2018 DHS wave, restricted to LGAs observed in all three waves. Column (3) omits the nighttime lights controls, as the DMSP-OLS series ends in 2013. Survey weights are de-normalized using updated population figures and numbers of interviewed women (Appendix A). Standard errors clustered at the LGA level are reported in parentheses and p-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Results for Diarrhea

Diarrhea	Full specification			Reduced specification		
	Original (1)	Reconstructed (2)	Extended (3)	Original (4)	Reconstructed (5)	Extended (6)
Forest loss						
This year	−0.607	−0.986	−0.419	−0.697	−1.084	−0.738
	(0.592)	(0.829)	(0.671)	(0.577)	(0.799)	(0.688)
	[0.305]	[0.235]	[0.533]	[0.227]	[0.175]	[0.283]
1 year ago	0.625	2.248**	−0.536	0.605	2.145**	−0.702
	(0.617)	(0.974)	(0.485)	(0.600)	(0.935)	(0.503)
	[0.311]	[0.021]	[0.269]	[0.313]	[0.022]	[0.163]
2 years ago	0.293	1.927	1.994**	0.442	2.183	2.355***
	(0.956)	(2.026)	(0.824)	(0.980)	(1.904)	(0.835)
	[0.760]	[0.342]	[0.016]	[0.652]	[0.252]	[0.005]
3 years ago	0.056	1.171	1.090	0.404	1.006	1.502
	(1.132)	(1.186)	(0.941)	(1.063)	(1.205)	(0.951)
	[0.960]	[0.324]	[0.247]	[0.704]	[0.404]	[0.114]
Observations	40,715	40,508	55,286	40,922	40,669	55,439
R^2	0.084	0.084	0.089	0.082	0.082	0.085

Notes: The dependent variable is the reported incidence of diarrhea in the two weeks preceding the survey among children under five. Columns (1)–(3) use the full specification of Berazneva and Byker (2017) (column 6 of their Table 1); columns (4)–(6) use the reduced specification described in Section 3.2, with the variables listed in Table A1. Columns (1) and (4) use the original analysis dataset, columns (2) and (5) the dataset reconstructed from raw sources (2008 and 2013 DHS waves), and columns (3) and (6) the extended dataset that adds the 2018 DHS wave, restricted to LGAs observed in all three waves. Column (3) omits the nighttime lights controls, as the DMSP-OLS series ends in 2013. Survey weights are de-normalized using updated population figures and numbers of interviewed women (Appendix A). Standard errors clustered at the LGA level are reported in parentheses and p-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Results for Respiratory Infections (Cough)

Respiratory (cough)	Full specification			Reduced specification		
	Original (1)	Reconstructed (2)	Extended (3)	Original (4)	Reconstructed (5)	Extended (6)
Forest loss						
This year	−1.193	−0.625	−1.012	−1.018	−0.106	−1.186
	(1.128)	(1.360)	(1.041)	(1.094)	(1.323)	(1.007)
	[0.290]	[0.646]	[0.331]	[0.352]	[0.936]	[0.239]
1 year ago	0.455	−0.033	1.310**	0.612	0.514	1.241*
	(0.779)	(1.134)	(0.657)	(0.772)	(1.164)	(0.645)
	[0.559]	[0.977]	[0.046]	[0.428]	[0.659]	[0.054]
2 years ago	−1.085	−3.547**	−1.122	−1.341	−4.351***	−1.155
	(0.835)	(1.672)	(1.204)	(0.846)	(1.644)	(1.192)
	[0.194]	[0.034]	[0.352]	[0.113]	[0.008]	[0.332]
3 years ago	0.515	2.512	0.438	0.359	1.973	0.591
	(2.266)	(2.513)	(1.509)	(2.155)	(2.504)	(1.500)
	[0.820]	[0.318]	[0.772]	[0.868]	[0.431]	[0.693]
Observations	40,618	40,407	55,198	40,823	40,568	55,351
R^2	0.093	0.092	0.088	0.091	0.090	0.086

Notes: The dependent variable is the reported incidence of cough in the two weeks preceding the survey among children under five. Columns (1)–(3) use the full specification of Berazneva and Byker (2017) (column 9 of their Table 1); columns (4)–(6) use the reduced specification described in Section 3.2, with the variables listed in Table A1. Columns (1) and (4) use the original analysis dataset, columns (2) and (5) the dataset reconstructed from raw sources (2008 and 2013 DHS waves), and columns (3) and (6) the extended dataset that adds the 2018 DHS wave, restricted to LGAs observed in all three waves. Column (3) omits the nighttime lights controls, as the DMSP-OLS series ends in 2013. Survey weights are de-normalized using updated population figures and numbers of interviewed women (Appendix A). Standard errors clustered at the LGA level are reported in parentheses and p-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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A Updated De-Normalization of Survey Weights

When pooling several DHS waves, Berazneva and Byker (2017) de-normalize the survey weights so that each wave is weighted according to the size of the population it represents. The replication code uses populations of women of reproductive age (WRA, aged 15–49) of 34,442,230 in 2008 and 39,172,540 in 2013, taken from an earlier revision of the UN World Population Prospects, and numbers of interviewed women of 33,385 in 2008 and 38,185 in 2013; the source cited in the code is no longer accessible. We update the population figures with estimates from the 2024 revision of the World Population Prospects for 2008 (35,882,027), 2013 (41,018,918), and 2018 (47,146,386) (United Nations, Department of Economic and Social Affairs, Population Division 2024). For the numbers of interviewed women, we use the figures from the DHS final reports: 33,385 in 2008, 38,948 in 2013, and 41,821 in 2018 (NPC and ICF, 2009, 2014, 2019). Following the original study, the de-normalized weight of observation i in survey year t is

$$w_{it}^{\mathrm{dn}} = \frac{v005_{it}}{1{,}000{,}000} \times \frac{\mathrm{PopWRA}_t}{N_t},$$

where $v005_{it}$ is the DHS women's sample weight (stored with six implied decimals), PopWRA_t is the UN estimate of the WRA population, and N_t is the number of women aged 15–49 interviewed in the respective DHS wave. Because the de-normalization multiplies all weights within a wave by the same factor, the update changes only the relative weight of the survey waves: an observation from 2008 receives 1.021 times the weight of a 2013 observation with the same DHS sample weight, compared with 1.006 in the original code. The update leaves the point estimates and standard errors essentially unchanged (compare column 1 of Table 4 with column 3 of Table 1 in Berazneva and Byker 2017). Details are provided in the do-files of our replication repository.

B Variables Included in Each Specification

Table A1: Variables Included in Each Specification

Variable	Description	Full	Reduced	Reason for exclusion
forest_loss_0	Forest loss in the survey year	Yes	Yes	
forest_loss_1–_3	Forest loss one, two, and three years before the survey	Yes	Yes	
i.DHSYEAR	Survey-wave fixed effects	Yes	Yes	
i.LGA	LGA fixed effects	Yes	Yes	
i.month#i.region#i.DHSYEAR	Month × region × survey-wave fixed effects	Yes	No	Correlated with
i.month	Month fixed effects	Yes	No	LGA and month
i.month#i.region	Month × region fixed effects	No	Yes	fixed effects
treecover_2000	Mean tree cover in 2000 (%)	Yes	Yes	
cec_ave_pt	Soil cation exchange capacity, average of 2.5 and 10 cm	Yes	No	
ph_ave_pt	Soil pH, average of 2.5 and 10 cm	Yes	No	Correlated with forest loss
occ_ave_pt	Soil organic carbon, average of 2.5 and 10 cm	Yes	No	
luminosity_chg_0–_2	Change in nighttime light coverage	Yes	No	
altitude	Altitude (m)	Yes	Yes	
no_HH_members	Number of household members (hv009)	Yes	Yes	
no_kids_under_5	Number of children under 5 in the household (hv014)	Yes	Yes	
time_to_water	Minutes to nearest water source (hv204)	Yes	Yes	
HH_head_edu_years	Years of education of household head (hv108_01)	Yes	Yes	
head_HH_age	Age of household head in years (hv220)	Yes	No	Not relevant
toilet	1 if household has a flush toilet (hv205)	Yes	No	Correlated with
firewood	1 if cooking fuel is firewood or straw (hv226)	Yes	No	rural residence and poverty
floor	1 if floor is earth, sand, or dung (hv213)	Yes	No	
rural	1 if rural (hv025)	Yes	Yes	
poorest	1 if in poorest wealth quintile (hv270)	Yes	Yes	
age	Child's current age (b8)	Yes	Yes	
age_resp	Age of respondent in years (v012)	Yes	No	Not relevant
edu_years	Years of education of respondent (v133)	Yes	No	
no_child_total	Number of children ever born (v201)	Yes	No	
no_child_living	Number of living children (v218)	Yes	No	Correlated with rural residence and poverty
resp_slept_net	1 if respondent slept under a bednet (v461)	Yes	No	
resp_works	1 if respondent works (v714)	Yes	No	
livewith	Number of children living with respondent (v202 + v203)	Yes	Yes	
christian	1 if respondent is Christian (v130)	Yes	Yes	
muslim	1 if respondent is Muslim (v130)	Yes	Yes	
yoruba	1 if respondent is Yoruba (v131)	Yes	Yes	
igbo	1 if respondent is Igbo (v131)	Yes	Yes	
hausa	1 if respondent is Hausa (v131)	Yes	Yes	
married	1 if respondent is married (v501)	Yes	Yes	
pregnant	1 if respondent is pregnant at the time of interview (v213)	Yes	Yes	

Notes: Variables included in the full specification of Berazneva and Byker (2017) (Equation 1) and in our reduced specification (Section 3.2). Indicator variables equal 1 if the stated condition holds and 0 otherwise; DHS variable codes are given in parentheses. The last column gives the reason for excluding a variable from the reduced specification.